Input-Contract-Aware Conditioning for Reproducible Volumetric Mapping Across Heterogeneous Robotic Platforms

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***Abstract*—Volumetric mapping pipelines rely on assumptions regarding pose validity, frame interpretation, cloud-pose timing, and observation admissibility before integration. In practice, these assumptions are rarely made explicit, making mapping failures difficult to attribute and replay conditions difficult to reproduce across heterogeneous robotic platforms and data sources. This paper presents an explicit and auditable input-contract formulation that supports reproducible replay conditions together with a backend-preserving wrapper that enforces the contract while leaving the underlying mapping pipeline unchanged. The approach is evaluated through paired replay experiments on Spot, ANYmal, and a retained Dynablox data source using archived PCD-derived proxy metrics and a runtime timing-admissibility audit. Results show small changes in the map-proxy metrics under already-compatible input conditions. When these conditions are violated, the wrapper makes the resulting differences observable and attributable to upstream input assumptions. These findings support input-contract-aware conditioning as a practical mechanism for making mapping assumptions visible, checkable, and explicitly controlled during replay.**

***Index Terms*—volumetric mapping, input conditioning, TSDF, sensor synchronization, heterogeneous robotic platforms**

I. Introduction

Within the field of robotic perception, various technologies have been developed with the aim of providing robots with a representation of their surroundings that enables navigation, exploration, and interaction tasks [1]. Among these technologies, dense volumetric mapping has become a mature solution for three-dimensional reconstruction and planning, relying on representations such as Truncated Signed Distance Fields (TSDF) and Euclidean Signed Distance Fields (ESDF) [2], [3]. In addition, dense volumetric mapping systems are commonly analyzed through the behavior of their fusion strategy, spatial representation, and consistency mechanisms. Consequently, much of the development of systems such as Voxblox [2], VDBFusion [4], DB-TSDF [5], and Voxgraph [6] has focused primarily on the behavior of the mapping backend.

Nevertheless, successful volumetric integration relies on a set of assumptions regarding the observations delivered to the backend. Typically, these systems assume that observations are already available in a usable form. Their performance therefore depends on the availability of a valid and sufficiently accurate pose estimate [2], on the consistency of reference-frame interpretations, on odometry satisfying geometric constraints such as gravity alignment [6], and on processes such as filtering, preprocessing, and motion undistortion being resolved prior to integration [4]. Similarly, LiDAR returns and observations originating from multiple sensors are typically transformed using pose estimates before being incorporated into the map [5].

Although these requirements are fundamental to successful integration, they are rarely treated as an explicit object of analysis [1]. Instead, many of them remain embedded within robot-specific calibration procedures, transform-tree definitions, synchronization mechanisms, and execution pipelines [7]. Consequently, when mapping behavior degrades, it is often difficult to determine whether the source of the problem lies in the volumetric backend itself or in the assumptions governing the observations delivered to it.

This challenge becomes more pronounced in heterogeneous robotic systems, where differences in sensing, calibration,

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frame conventions, and runtime conditions can alter the observations presented to the same backend [8]–[11]. As a result, changes in mapping behavior may originate from differences in upstream input conditions rather than from the volumetric integration process itself.

Motivated by these observations, this paper investigates the input boundary of volumetric mapping through an input-contract-aware conditioning architecture for heterogeneous robotic platforms and data sources. The main contribution of this work is an explicit and auditable input-contract formulation for controlled replay that externalizes assumptions regarding pose validity, frame interpretation, timing consistency, and observation admissibility [12]. Unlike conventional preprocessing, the proposed contract is treated as an experimental boundary whose admission and interpretation decisions are logged during replay, allowing map differences to be analyzed from input conditions while preserving the DB-TSDF backend pipeline [5] unchanged.

As secondary contributions, the paper also presents:

- A backend-preserving conditioning architecture that enforces the proposed contract prior to volumetric integration while remaining decoupled from the underlying TSDF implementation.
- A cross-platform evaluation showing that explicit input contracts enable attribution of mapping variability to upstream assumptions while producing only small map-proxy changes under already-compatible input conditions.

II. Related Work

A. Volumetric Mapping Backends

Volumetric mapping has matured around signed-distance-field representations for dense geometric reconstruction and planning. Representative examples include TSDF/ESDF-based systems such as Voxblox [2], sparse volumetric fusion methods such as VDBFusion [4], and GPU-accelerated implementations such as nvblox [3]. Other works have explored global consistency, elastic updates, and semantic extensions on top of the same general volumetric paradigm [6], [13]–[16]. Across these systems, a common assumption is that incoming observations are already admissible for volumetric integration.

In the present work, DB-TSDF is retained as the mapping backend [5]. Rather than proposing a new fusion engine, we deliberately preserve the backend and focus on the conditions governing the observations delivered to it. This design allows mapping differences to be interpreted in terms of input admissibility rather than backend modification.

B. Timing, Frames, and Reproducible Robotic Mapping

A parallel line of work has shown that mapping performance depends strongly on upstream conditions such as calibration fidelity, frame correctness, and timing consistency. Underwood et al. showed that accurate mapping depends on the quality of the sensor-to-map transform chain [8], while Furgale et al. and Qiu et al. demonstrated the importance of joint spatial-temporal calibration and temporal consistency in heterogeneous sensor systems [9], [17]. Timing-sensitive effects have also been studied directly in SLAM systems, where software timing variations can degrade performance even when the mapping formulation itself remains unchanged [18].

Collectively, these studies show that volumetric backends, calibration methods, and synchronization strategies are all well established. However, they are usually treated as separate concerns, backend design on one side, and sensing, timing, and frame handling on the other. Existing literature reveals that pose validity, frame interpretation, timing consistency, and observation admissibility have rarely been formalized as components of an explicit and enforceable input contract governing the observations delivered to a volumetric mapping backend.

III. Methods

Figure 1 summarizes the workflow from heterogeneous inputs through contract enforcement to backend-preserving conditioning and map-proxy evaluation.

A. Observation Model

A volumetric mapping backend implicitly assumes that incoming observations are expressed under compatible pose, frame, timing, and observation-quality conventions. When these assumptions differ across robotic platforms, the same backend may produce substantially different mapping outcomes for reasons that originate upstream of volumetric integration.

We represent each observation as defined by Eq. 1.

$$z_t = \mathcal{P}_t, {}^{W}T_{S,t}, f_t, \tau_t, \tag{1}$$

where $\mathcal{P}_t$ denotes the point cloud, ${}^{W}T_{S,t}$ the sensor pose used for integration, f_t the relevant frame convention, and τ_t the observation timestamp. The frame term f_t abstracts the frame-resolution and transform-interpretation policies used to express observations in a common integration frame.

B. Input Contract

Rather than treating platform assumptions as implicit preprocessing requirements, the proposed approach elevates them to explicit admissibility conditions that can be enforced, audited, and compared across heterogeneous robotic systems.

The input contract is defined through the admissibility function, as given by Eq. 2.

$$a_t = \mathcal{A}(\mathcal{P}_t, {}^{W}T_{S,t}, f_t, \tau_t) \in \{0, 1\}, \tag{2}$$

equation 3 determines whether an observation is eligible for backend integration.

$$(\mathcal{P}_t, {}^{W}T_{S,t}) \rightarrow \mathcal{M}_{TSDF} \quad \text{only if} \quad a_t = 1, \tag{3}$$

where M_{TSDF} denotes the unchanged TSDF backend.

The admissibility decision is decomposed into four contract dimensions, as defined in Eq. 4.

$$a_t = a_t^{pose} \wedge a_t^{frame} \wedge a_t^{time} \wedge a_t^{obs}, \tag{4}$$

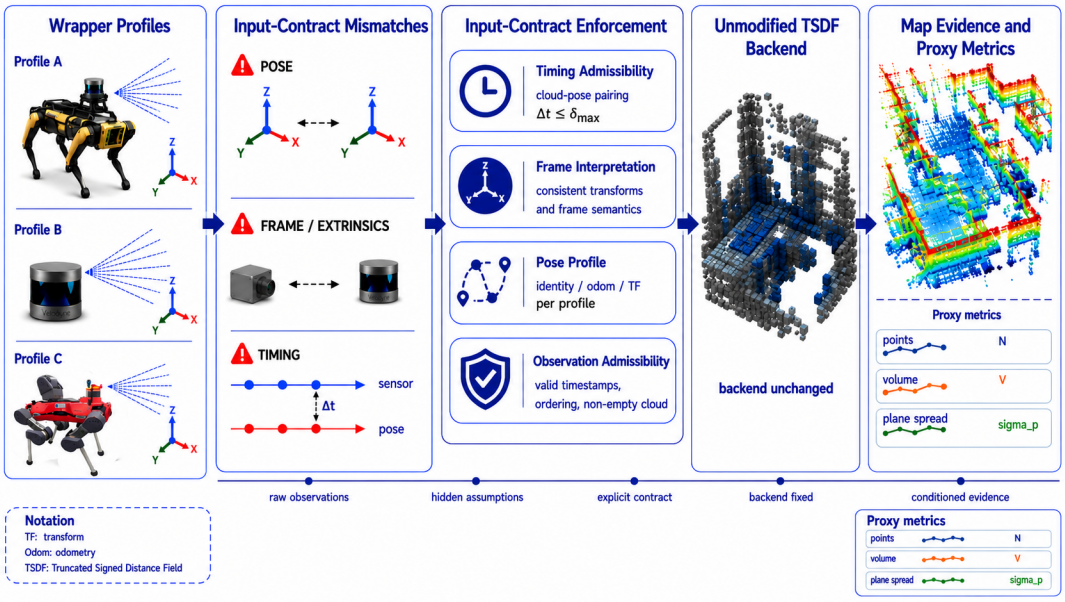


Fig. 1. Overview of the proposed input-contract-aware volumetric mapping framework. The left side illustrates the problem addressed in this work: heterogeneous robot and data profiles may introduce pose, frame, and timing mismatches before mapping. The central block represents the proposed contribution, namely an explicit input contract that enforces timing, frame, pose, and observation admissibility. Admissible observations are then forwarded to an unchanged DB-TSDF backend. Finally, the resulting maps are analyzed through map-level proxy metrics, providing evidence of the effects of input conditioning without modifying the backend.

where a_t^{pose}, a_t^{frame}, a_t^{time}, and a_t^{obs} denote pose validity, frame admissibility, timing admissibility, and observation-quality admissibility, respectively. Their operational meaning in the retained experiments is summarized in Table I.

TABLE I
OPERATIONAL DIMENSIONS OF THE RETAINED INPUT CONTRACT.

Dimension	What is checked	Action
Pose	valid pose sample for cloud pairing	accept/reject
Timing	cloud-pose skew below 50 ms	accept/reject
Observation	valid timestamp, causal order, non-empty cloud	accept/reject
Frame	coherent interpretation under the selected profile	normalize/interpret

Frame admissibility evaluates whether an observation can be consistently interpreted within the selected integration-frame convention before backend integration. Thus, timing admissibility, pose availability, timestamp validity, ordering, and empty-cloud checks are enforced operationally during replay, while frame admissibility is realized through an externally configured pose/transform profile; the wrapper neither infers the correct profile nor estimates missing extrinsics.

This formulation preserves the volumetric backend unchanged while relocating platform-dependent assumptions to an explicit wrapper layer. Consequently, backend behavior is evaluated under a common admissibility model rather than through platform-specific preprocessing assumptions

C. Timing Admissibility

For timing consistency, each cloud timestamp τ_t^P is associated with the most recent pose sample available before cloud acquisition, as defined in Eq. 5.

$$i^* = \arg\max_i \{\tau_i^T \mid \tau_i^T \leq \tau_t^P\}, \tag{5}$$

The corresponding temporal skew is computed as defined in Eq. 6.

$$\Delta\tau_t = \left|\tau_t^P - \tau_{i^*}^T\right|. \tag{6}$$

Timing admissibility is then defined by Eq. 7.

$$a_t^{time} = \mathbb{I}\left[\Delta\tau_t \leq \delta_{\max}\right], \tag{7}$$

where $\delta_{\max}$ denotes the maximum allowable cloud-pose skew. The retained experiments used $\delta_{\max} = 0.05$ s as an operational threshold, not as a platform-independent optimum.

D. Wrapper Profiles

The architecture keeps platform-specific assumptions outside the mapping backend through profile-dependent pose and frame interpretation, as given by Eq. 8.

$${}^W T_{S,t} = \begin{cases} \mathbf{I}, & \text{identity sensor pose,} \\ {}^W\hat{T}_{S,t}, & \text{normalized sensor pose,} \\ {}^W T_{B,t}\,{}^B T_S, & \text{body pose with extrinsic.} \end{cases} \tag{8}$$

where W denotes the mapping frame, B the robot body frame, and S the mapping sensor frame. The first case assumes an identity sensor pose, the second uses a provided sensor-frame estimate, and the third composes body odometry with a static sensor extrinsic.

These profiles allow heterogeneous robotic platforms to satisfy a common input contract while preserving the underlying

TSDF backend unchanged. As a result, backend behavior can be evaluated under comparable observation semantics despite differences in sensing, localization, and frame conventions.

E. Backend-Preserving Conditioning

The dense mapping path receives a conditioned cloud-pose stream and integrates it through the unchanged DB-TSDF backend. The wrapper is responsible for input normalization and admissibility checks, while the backend remains isolated from platform-specific pose and frame handling. The conditioned observation stream is expressed by Eq. 9.

$$\mathcal{Z}^{+} = z_t \mid \mathcal{A}(z_t) = 1, \tag{9}$$

where $\mathcal{Z}^{+}$ denotes the subset of observations that satisfy the input contract and are therefore eligible for TSDF integration.

Because the volumetric backend remains unchanged across all experiments, observed differences in the resulting maps can be associated with variations in input admissibility and conditioning rather than with modifications of the TSDF integration process itself. The evaluation therefore relies on map-level proxy metrics to characterize the observable effects of input conditioning.

F. Map-Proxy Metrics

Since the compared datasets do not share a common geometric ground truth, the evaluation relies on PCD-derived proxy metrics computed offline from exported map clouds under fixed backend settings.

An exported map cloud is defined as in Eq. 10.

$$\mathcal{M} = \{\mathbf{p}_i\}_{i=1}^{N}, \tag{10}$$

where $\mathbf{p}_i$ is the ith map point.

While the point count is given by Eq. 11, providing a proxy for the amount of retained geometric evidence:

$$N = |\mathcal{M}|. \tag{11}$$

The bounding-box volume is defined as in Eq. 12, and measures global three-dimensional map spread:

$$V_{bbox} = \prod_{d \in x,y,z} (\max_i p_{i,d} - \min_i p_{i,d}). \tag{12}$$

The horizontal footprint is defined as in Eq. 13, and summarizes planar compactness:

$$A_{xy} = \prod_{d \in x,y} (\max_i p_{i,d} - \min_i p_{i,d}), \tag{13}$$

where $p_{i,d}$ denotes the coordinate of point $\mathbf{p}_i$ along dimension d. Finally, the point density is defined as in Eq. 14:

$$\rho = \frac{N}{V_{bbox}}. \tag{14}$$

The plane-spread proxy σ_p corresponds to the standard deviation of point-to-plane residuals after least-squares fitting on a low-lying support subset selected from the lower 20% of the map vertical extent.

Taken together, N, V_{bbox}, A_{xy}, ρ, and σ_p characterize retained geometric information, global spread, planar compactness, volumetric concentration, and local surface dispersion, respectively; they are diagnostic descriptors rather than measures of geometric reconstruction accuracy.

IV. Experiments and Results

A. Data Sources and Reproducible Execution

The experimental protocol adopts a common replay boundary in which archived sensor streams are reprocessed under fixed backend settings and explicit wrapper profiles. Spot recordings and the retained Dynablox sequence were replayed through reproducible `rosbag2` execution. The Dynablox case uses a public LiDAR dataset originally released for dynamic-scene mapping evaluation [19].

ANYmal required an additional normalization step because the retained recording did not originate from the same replay pipeline. The sequence was therefore converted into the common `rosbag2`-based execution environment and evaluated under the same frozen backend configuration used throughout the study.

B. Cross-Platform Input-Contract Evaluation

Figure 2 illustrates representative paired maps under fixed backend settings. Spot and Dynablox compare direct DB-TSDF execution against the wrapper-mediated path, while ANYmal is evaluated through a wrapper-profile ablation between odometry-only integration and odometry composed with a static LiDAR extrinsic. Map-proxy metrics are computed from exported PCD maps following Section III-F; Table II reports the archived values, and Table III summarizes paired changes.

The results show small paired differences for Spot and Dynablox, as quantified in Table III, whereas ANYmal changes substantially when the static LiDAR extrinsic is incorporated into the pose interpretation path. These comparisons indicate that heterogeneous platforms and data sources can present different input regimes to the same backend despite an unchanged integration configuration.

TABLE II
ARCHIVED MAP-PROXY VALUES FOR PAIRED RUNS.

Source	Condition	N (10^6 pts)	V_{bbox} (10^3 m^3)	ρ (pts/m^3)	A_{xy} (m^2)	σ_p (m)
Spot	direct	2.471	20.33	121.5	2363.60	0.1931
Spot	identity wrapper	2.428	20.09	120.9	2363.60	0.1918
ANYmal	odometry only	9.952	53.39	186.4	3559.38	0.3206
ANYmal	odometry + extrinsic	2.625	32.11	81.8	2396.27	0.3462
Dynablox	direct	4.737	61.25	77.3	3093.24	0.2404
Dynablox	TF-normalized pose	4.671	61.48	76.0	3105.16	0.2379

C. Runtime Timing/Admissibility Audit

The wrapper emits runtime status information exposing timing and admissibility conditions during replay. Table IV summarizes contract-level quantities including admission ratio,

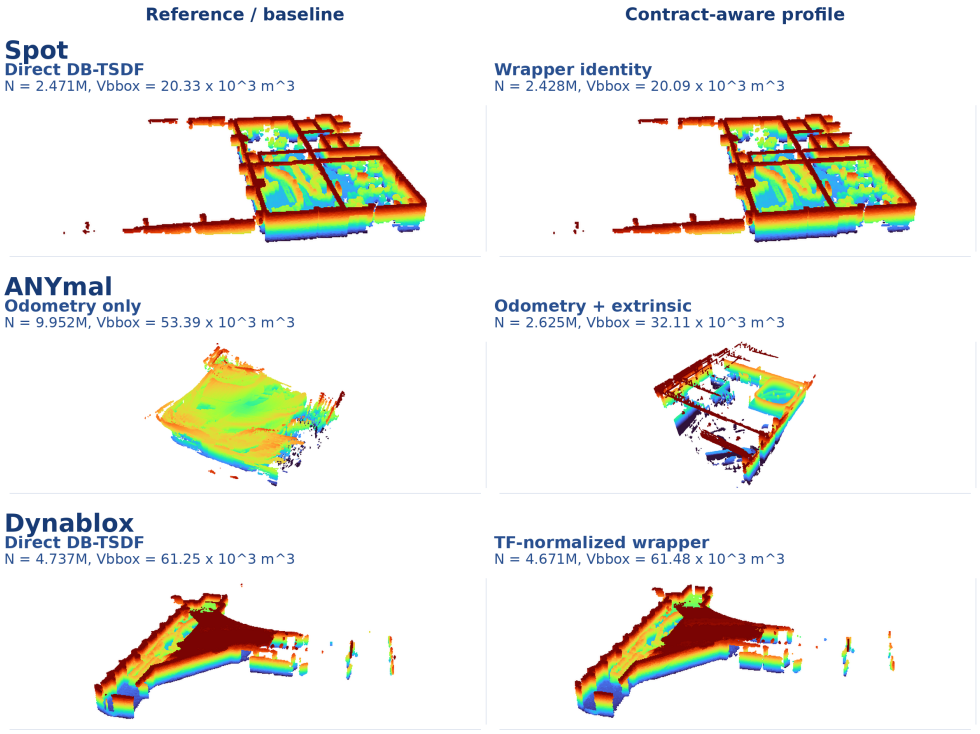


Fig. 2. Cross-platform profile comparison under a fixed backend. Spot and Dynablox pair direct DB-TSDF baselines with wrapper-mediated input conditioning, while ANYmal compares wrapper profiles with and without the static LiDAR extrinsic. Within each paired comparison, the visualizations use the same viewpoint and a common height-color convention.

TABLE III
PAIRED MAP-PROXY CHANGES BY PLATFORM.

Source	Paired change	ΔN	ΔV	$\Delta\rho$	ΔA_{xy}	$\Delta\sigma_p$
Spot	direct → identity	-1.7%	-1.2%	-0.6%	0.0%	-0.7%
ANYmal	odom → extr.	-73.6%	-39.9%	-56.2%	-32.7%	+8.0%
Dynablox	direct → TF pose	-1.4%	+0.4%	-1.8%	+0.4%	-1.0%

missing-pose rejections, pose-skew rejections, and cloud-pose skew statistics.

Rather than estimating source-level synchronization accuracy, the audit reports wrapper-level admissibility observables during replay. Spot remained highly admissible, whereas the retained ANYmal replays exhibited lower admission ratios through runtime missing-pose rejections; pose-skew rejections remained null.

TABLE IV
TIMING/ADMISSIBILITY AUDIT FROM WRAPPER LOGS.

Audit stream	Clouds	Acc. ratio	Miss. pose	Skew rej.	Skew (ms) mean / max
Spot dynamic	359	0.978	8	0	5.2/25.3
Spot static	475	0.973	13	0	11.2/24.9
ANYmal odom-only	1153	0.846	178	0	1.6/38.7
ANYmal odom+extr.	1153	0.883	135	0	1.6/27.6

D. Controlled Contract-Violation Replay

To complement the cross-platform evaluation with a controlled within-sequence comparison, the retained Spot replay was repeated under a fixed backend while reducing upstream pose availability through pose-stream decimation. The baseline used the original odometry stream, while the controlled conditions forwarded every second, fourth, or eighth pose message under the same admissibility parameters. Table V shows that as pose sparsity increased, the accepted ratio dropped from 0.993 to 0.682 and 0.364, while pose-skew rejections increased, making the induced timing violations observable at the admissibility boundary under an unchanged backend.

TABLE V
CONTROLLED SPOT POSE-DECIMATION REPLAY UNDER A FIXED BACKEND.

Condition	Acc. ratio	Skew rej.	N (10^6)	V_{bbox} (m^3)
Baseline	0.993	0	2.463	20208.8
Mild decimation (every 2nd pose)	0.993	0	2.422	20090.6
Medium decimation (every 4th pose)	0.682	88	2.187	20119.3
Strong decimation (every 8th pose)	0.364	178	1.607	19356.4

V. DISCUSSION

The results suggest that pose validity, frame interpretation, timing consistency, and observation admissibility constitute an important source of variability in volumetric mapping systems. This observation is consistent with the broader SLAM literature, where sensing, calibration, synchronization, and estimation assumptions frequently remain outside the backend formulation itself [1], [9], [18].

The effect of explicit contract enforcement was not uniform across platforms. Spot and Dynablox exhibited only small

changes between paired executions, suggesting that the retained input regimes already satisfied most backend assumptions. In these cases, the limited proxy variations indicate that explicit contract enforcement introduced only small observable changes when the underlying conditions were already compatible with the mapping pipeline. This interpretation is further supported by the controlled Spot pose-decimation sweep, where mild pose sparsity remained admissible, whereas stronger upstream decimation produced explicit admissibility failures while the backend remained unchanged. This controlled result shows that violations of the declared input regime become visible and attributable at the wrapper boundary before volumetric integration.

In contrast, ANYmal exhibited substantially larger changes when pose interpretation included the static LiDAR extrinsic. This case is therefore interpreted as a frame-profile sensitivity experiment: the results show that pose and frame interpretation can materially alter the geometric stream presented to the same volumetric backend, but without an independent geometric reference they do not establish which reconstruction is more accurate, consistent with previous studies on calibration and extrinsic sensitivity in mapping systems [17], [20]. Making these assumptions explicit allows observed map differences to be associated with identifiable input conditions rather than being attributed solely to the mapping backend.

The runtime audit provides complementary evidence for the proposed input contract. The audit exposes cloud admission, pose availability, and cloud-pose timing constraints during replay. In particular, the ANYmal streams show profile-dependent missing-pose rejections even when pose-skew rejections remain null. As a result, integration conditions become explicit, auditable, and consistently controlled during replay rather than remaining implicit within the mapping pipeline. The reported results should nevertheless be interpreted as evidence of input-regime sensitivity under heterogeneous datasets and platform-specific paired comparisons rather than as ground-truth reconstruction-accuracy evaluations.

VI. Conclusion

This paper presented an input-contract-aware conditioning architecture for volumetric mapping that makes pose, frame, timing, and admissibility assumptions explicit before integration while preserving the underlying TSDF backend. Experimental results showed that different input regimes can produce different mapping outcomes under an unchanged backend configuration, while the runtime audit demonstrated that contract-level conditions remain observable during replay. The combined experimental and audit evidence indicates that input assumptions constitute an identifiable and auditable source of mapping variability under the retained backend configuration. Future work will investigate how the proposed input-contract boundary can support higher-level world-modeling capabilities, including semantic scene representations, temporal entity reasoning, and multi-robot mapping.

References

[1] C. Cadena, L. Carlone, H. Carrillo, Y. Latif, D. Scaramuzza, J. Neira, I. Reid, and J. J. Leonard, "Past, present, and future of simultaneous localization and mapping: Toward the robust-perception age," *IEEE Transactions on Robotics*, vol. 32, no. 6, pp. 1309–1332, 2016.

[2] H. Oleynikova, Z. Taylor, M. Fehr, R. Siegwart, and J. Nieto, "Voxblox: Incremental 3d euclidean signed distance fields for on-board mav planning," in *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2017, pp. 1366–1373.

[3] A. Millane *et al.*, "nvblox: Gpu-accelerated incremental signed distance field mapping," in *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 2698–2705.

[4] I. Vizzo, T. Guadagnino, J. Behley, and C. Stachniss, "VDBfusion: Flexible and efficient TSDF integration of range sensor data," *Sensors*, vol. 22, no. 3, 2022.

[5] J. E. Maese, L. Merino, and F. Caballero, "Db-tsdf: Directional bitmask-based truncated signed distance fields for efficient volumetric mapping," in *2026 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2026, pp. 21 206–21 212.

[6] V. Reijgwart, A. Millane, H. Oleynikova, R. Siegwart, C. Cadena, and J. I. Nieto, "Voxgraph: Globally consistent, volumetric mapping using signed distance function submaps," *IEEE Robotics and Automation Letters*, vol. 5, no. 1, pp. 227–234, 2020.

[7] K. Ebadi, L. Bernreiter, H. Biggie, G. Catt, Y. Chang, A. Chatterjee, C. Denniston, S.-P. Deschênes, K. Harlow, S. Khattak *et al.*, "Present and future of slam in extreme environments: The darpa subt challenge," *IEEE Transactions on Robotics*, vol. 40, pp. 936–959, 2024.

[8] J. Underwood, A. Hill, T. Peynot, and S. Scheding, "Error modeling and calibration of exteroceptive sensors for accurate mapping applications," *Journal of Field Robotics*, vol. 27, no. 1, pp. 2–20, 2010.

[9] K. Qiu, T. Qin, J. Pan, S. Liu, and S. Shen, "Real-time temporal and rotational calibration of heterogeneous sensors using motion correlation analysis," *IEEE Transactions on Robotics*, vol. 37, no. 2, pp. 587–602, 2021.

[10] Y. Chang *et al.*, "Lamp 2.0: A robust multi-robot slam system for operation in challenging large-scale underground environments," *IEEE Robotics and Automation Letters*, vol. 7, no. 4, pp. 9175–9182, 2022.

[11] M. Tranzatto *et al.*, "Cerberus in the darpa subterranean challenge," *Science Robotics*, vol. 7, no. 66, 2022.

[12] S. M. Castaño Vanegas, "Input-contract-aware, source code, datasets and recorded runs," GitHub repository, 2026, accessed on September 22, 2026. [Online]. Available: https://github.com/EESC-LabRoM/volumetric-mapping-input-contract/tree/main

[13] Y. Wang *et al.*, "Elastic and efficient LiDAR reconstruction for large-scale exploration tasks," in *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 2021, pp. 5035–5041.

[14] J. McCormac, A. Handa, A. Davison, and S. Leutenegger, "Semanticfusion: Dense 3d semantic mapping with convolutional neural networks," in *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 2017, pp. 4628–4635.

[15] G. Narita, T. Seno, T. Ishikawa, and Y. Kaji, "Panopticfusion: Online volumetric semantic mapping at the level of stuff and things," in *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2019, pp. 4205–4212.

[16] L. Schmid *et al.*, "Panoptic multi-tsdfs: A flexible representation for online multi-resolution volumetric mapping and long-term dynamic scene consistency," in *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, 2022, pp. 8018–8024.

[17] P. Furgale, J. Rehder, and R. Siegwart, "Unified temporal and spatial calibration for multi-sensor systems," in *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2013, pp. 1280–1286.

[18] A. Li, H. Liu, J. Wang, and N. Zhang, "From timing variations to performance degradation: Understanding and mitigating the impact of software execution timing in SLAM," in *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022, pp. 13 308–13 315.

[19] Autonomous Systems Lab, ETH Zurich, "Dynablox dataset," https://projects.asl.ethz.ch/datasets/dynablox/, accessed: Jun. 5, 2026.

[20] J. Jiao, Y. Yu, Q. Liao, H. Ye, R. Fan, and M. Liu, "Automatic calibration of multiple 3d lidars in urban environments," in *2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2019, pp. 15–20.